

Qualification Pack

Scientific Assistant in Data Science for Life Sciences

QP Code: NIE/SSC/Q1701

Version: 1.0

NSQF Level: 4

National Institute of Electronics and Information Technology (NIELIT) || NIELIT Bhawan, Plot No. 3, PSP
Pocket, Sector-8, Dwarka
New Delhi-110077 || email:puneetnema@nielit.gov.in

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Qualification Pack

NIE/SSC/Q1701: Scientific Assistant in Data Science for Life Sciences

Brief Job Description

This Short-term Certificate Course is targeted at creating qualified professionals in the field of Bioinformatics, which will help in the employment or Entrepreneurship development of the qualifier. The purpose of this qualification is to train the students of Life Sciences with the basics of Bioinformatics to skill them and increase their employability in the field of Life Sciences and allied disciplines.

Personal Attributes

The required personal attributes span a broad spectrum essential for success in bioinformatics and data security law. An analytical mindset and adept problem-solving skills are fundamental for understanding algorithms and utilizing bioinformatics tools effectively. Proficiency in Python programming, coupled with adaptability, enables individuals to address diverse biological challenges with versatility. Attention to detail, methodical analysis, and a commitment to compliance standards are vital for comprehending biomolecular structures and ensuring data security law adherence.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [NIE/SSC/N1801: Algorithmic perspective of Bioinformatics and introduction to Bioinformatics tools.](#)
2. [NIE/SSC/N1802: Solving biological problems using Python programming](#)
3. [NIE/SSC/N1803: Analysis and understanding of biomolecular structures using Bioinformatics tools](#)
4. [NIE/SSC/N1804: Data security law](#)
5. [NIE/SSC/N1810: Implementation of project on bioinformatics](#)
6. [DGT/VSQ/N0102: Employability Skills \(60 Hours\)](#)

Qualification Pack (QP) Parameters

Sector	IT-ITeS
Sub-Sector	Artificial Intelligence
Occupation	Bioinformatics Instrumentation
Country	India

Qualification Pack

NSQF Level	4
Credits	15
Aligned to NCO/ISCO/ISIC Code	4
Minimum Educational Qualification & Experience	12th grade Pass (12th Grade Pass with biology as mandatory subject) OR 12th Class (* Biology should be the mandatory subject in class 12th or Equivalent and having knowledge of data crunching)
Minimum Level of Education for Training in School	
Pre-Requisite License or Training	NA
Minimum Job Entry Age	16 Years
Last Reviewed On	NA
Next Review Date	15/03/2027
NSQC Approval Date	15/03/2024
Version	1.0
Reference code on NQR	QG-04-IT-02180-2024-V1-NIELIT
NQR Version	1

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NIE/SSC/N1801: Algorithmic perspective of Bioinformatics and introduction to Bioinformatics tools.

Description

This NOS delves into bioinformatics from algorithmic and tool-based perspectives. It covers basic biology, molecular biology concepts, and DNA technology. Students learn about biological data formats, databases, and sequence alignment techniques. The course also explores gene prediction, molecular phylogeny, pattern matching, and pathways/systems biology analysis.

Scope

The scope covers the following :

- Algorithmic perspectives and introduction to bioinformatics tools.
- Basics of molecular biology, recombinant DNA technology, and biological data file formats.
- Introduction to biological databases, sequence alignment techniques, gene prediction methods, and molecular phylogeny.
- Exploring pattern matching, pathways/systems biology analysis, and simulation of cellular activities.

Elements and Performance Criteria

Foundational Concepts

To be competent, the user/individual on the job must be able to:

- PC1.** Demonstrate the ability to apply algorithmic thinking in solving bioinformatics problems, such as sequence alignment or motif discovery.
- PC2.** Utilize bioinformatics tools effectively to analyze biological data, including sequence data and structural information.
- PC3.** Explain the principles of molecular biology and DNA technology and their relevance to bioinformatics applications, such as gene prediction or protein structure analysis.

Data Handling and Analysis

To be competent, the user/individual on the job must be able to:

- PC4.** Successfully navigate and work with different biological data file formats (GenBank, FASTA, etc.) using appropriate conversion tools.
- PC5.** Effectively use databases like NCBI or EMBL to retrieve and analyze biological data for specific research questions or projects.

Advanced Applications

To be competent, the user/individual on the job must be able to:

- PC6.** Apply pattern matching techniques to identify regulatory sequences or motifs within DNA sequences, demonstrating an understanding of motif discovery algorithms.
- PC7.** Analyze biological pathways and systems, including metabolic networks, and conduct control analysis or simulation studies to understand cellular activities.
- PC8.** Use simulation methods to model and analyze biological processes, such as protein-protein interactions or molecular dynamics, and interpret the results in the context of biological function.

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Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.**
- Knowledge of Bioinformatics Tools and Algorithms:
 - Understanding of bioinformatics tools for sequence analysis, motif discovery, and phylogenetic analysis.
 - Familiarity with algorithms such as sequence alignment (Needleman-Wunsch, Smith Waterman), gene prediction methods, and phylogenetic tree construction.
 - Ability to apply heuristic methods like FASTA and BLAST for sequence similarity searches and motif identification.
- KU2.**
- Knowledge of Biological Data Handling:
 - Understanding of different file formats used in bioinformatics (GenBank, FASTA, etc.) and their conversion tools.
 - Knowledge of biological databases like NCBI, EMBL, and their classification systems.
 - Ability to handle and interpret biological data from databases and files for analysis and research purposes.
- KU3.**
- Understanding of Molecular Biology Concepts:
 - Knowledge of basic molecular biology principles, including DNA technology, gene prediction, and protein structure.
 - Understanding of molecular phylogeny, evolutionary models, and methods for constructing phylogenetic trees.
 - Familiarity with pathways and systems biology, including metabolic network analysis and simulation techniques.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.**
- Analytical Thinking:
 - Ability to analyze biological data using bioinformatics tools and algorithms.
 - Critical thinking skills in interpreting sequence alignments, gene predictions, and phylogenetic trees.
 - Capability to identify patterns, motifs, and regulatory sequences in DNA sequences.
- GS2.**
- Research Skills:
 - Proficiency in navigating and utilizing biological databases for research purposes.
 - Skill in conducting literature reviews and staying updated with advancements in bioinformatics.
 - Ability to apply computational methods to solve biological problems and contribute to scientific research.
- GS3.**
- Communication and Presentation:
 - Effective communication of bioinformatics analyses and findings to peers and stakeholders.
 - Skill in presenting complex biological data, algorithms, and models in a clear and concise manner.
 - Capability to collaborate with interdisciplinary teams and convey bioinformatics concepts to diverse audiences.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Foundational Concepts</i>	-	-	-	-
PC1. Demonstrate the ability to apply algorithmic thinking in solving bioinformatics problems, such as sequence alignment or motif discovery.	-	-	-	-
PC2. Utilize bioinformatics tools effectively to analyze biological data, including sequence data and structural information.	-	-	-	-
PC3. Explain the principles of molecular biology and DNA technology and their relevance to bioinformatics applications, such as gene prediction or protein structure analysis.	-	-	-	-
<i>Data Handling and Analysis</i>	-	-	-	-
PC4. Successfully navigate and work with different biological data file formats (GenBank, FASTA, etc.) using appropriate conversion tools.	-	-	-	-
PC5. Effectively use databases like NCBI or EMBL to retrieve and analyze biological data for specific research questions or projects.	-	-	-	-
<i>Advanced Applications</i>	-	-	-	-
PC6. Apply pattern matching techniques to identify regulatory sequences or motifs within DNA sequences, demonstrating an understanding of motif discovery algorithms.	-	-	-	-
PC7. Analyze biological pathways and systems, including metabolic networks, and conduct control analysis or simulation studies to understand cellular activities.	-	-	-	-
PC8. Use simulation methods to model and analyze biological processes, such as protein-protein interactions or molecular dynamics, and interpret the results in the context of biological function.	-	-	-	-
NOS Total	50	23	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	NIE/SSC/N1801
NOS Name	Algorithmic perspective of Bioinformatics and introduction to Bioinformatics tools.
Sector	IT-ITeS
Sub-Sector	
Occupation	Bioinformatics Data Analytics
NSQF Level	4
Credits	3
Version	1.0
Last Reviewed Date	15/03/2024
Next Review Date	15/03/2027
NSQC Clearance Date	15/03/2024

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NIE/SSC/N1802: Solving biological problems using Python programming

Description

The NOS delves into essential programming and computer fundamentals crucial for tackling biological challenges using Python. Students learn about computer systems, data representations, and IEEE standards, alongside scripting in Linux/Unix for bioinformatics tasks. Python programming basics cover input/output, data types, operators, and control structures, while emphasizing modular programming and functional decomposition. The course also introduces object-oriented programming concepts, aggregate data types, and fundamental data structures in Python, vital for efficient data handling and algorithmic design in bioinformatics.

Scope

The scope covers the following :

- Basic Computer Fundamentals: Understanding computer systems, data representation, and CPU organization.
- Scripting and Linux Commands: Learning basic scripting in Linux/Unix for bioinformatics tasks.
- Programming in Python: Covering Python basics, file handling, and modular programming concepts.
- Object-Oriented Programming (OOP) and Data Structures: Exploring OOP principles and essential data structures in Python for bioinformatics applications.

Elements and Performance Criteria

Computer Fundamentals

To be competent, the user/individual on the job must be able to:

- PC1.** Demonstrate understanding of number systems, including binary, decimal, and hexadecimal conversions.
- PC2.** Explain the functionality of CPU components such as ALU, registers, and memory in program execution.
- PC3.** Analyze and compare different data representations like integers, floating-point numbers, and character encoding.

Scripting and Linux Commands

To be competent, the user/individual on the job must be able to:

- PC4.** Create and execute basic scripts in Linux/Unix for file management and automation tasks.
- PC5.** Apply Linux commands effectively for navigating the file system, managing processes, and configuring system settings.
- PC6.** Troubleshoot scripting errors and utilize command-line utilities for system administration tasks.

Python Programming

To be competent, the user/individual on the job must be able to:

- PC7.** Develop Python programs that handle input/output operations, including reading from and writing to files.
- PC8.** Implement control structures like loops, conditions, and functions to solve programming challenges.

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PC9. Design object-oriented Python programs using classes, inheritance, and encapsulation principles for code organization and reusability

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.**
- Computer Fundamentals:
 - Knowledge of binary, decimal, and hexadecimal number systems and their conversions.
 - Understanding the role and function of CPU components like ALU, registers, and memory in program execution.
 - Familiarity with data representations such as integers, floating-point numbers, and character encoding.
- KU2.**
- Scripting and Linux Commands:
 - Understanding of basic scripting concepts and their application in Linux/Unix environments.
 - Knowledge of essential Linux commands for file management, process control, and system configuration.
 - Ability to troubleshoot scripting issues and utilize command-line tools effectively.
- KU3.**
- Python Programming:
 - Knowledge of Python syntax, data types, and basic programming constructs like loops and conditions.
 - Understanding of input/output operations in Python, including file handling and string processing.
 - Familiarity with object-oriented programming concepts such as classes, inheritance, and polymorphism in Python.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.**
- Problem-Solving Skills:
 - Ability to analyze biological problems and translate them into Python programming solutions.
 - Proficiency in using programming concepts and techniques to develop efficient and effective solutions.
 - Capability to debug code, identify errors, and implement appropriate modifications for problem resolution.
- GS2.**
- Technical Skills:
 - Competence in using computer systems and understanding fundamental computing concepts.
 - Proficiency in writing scripts and executing Linux commands for data analysis and manipulation.
 - Skill in programming with Python, including data handling, algorithm implementation, and software development practices.
- GS3.**
- Analytical Thinking:
 - Capability to assess complex biological data and formulate algorithms for data processing and interpretation.
 - Proficiency in analyzing requirements, designing logical solutions, and implementing them using Python programming.
 - Ability to evaluate program performance, optimize code, and apply best practices in software development.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Computer Fundamentals</i>	-	-	-	-
PC1. Demonstrate understanding of number systems, including binary, decimal, and hexadecimal conversions.	-	-	-	-
PC2. Explain the functionality of CPU components such as ALU, registers, and memory in program execution.	-	-	-	-
PC3. Analyze and compare different data representations like integers, floating-point numbers, and character encoding.	-	-	-	-
<i>Scripting and Linux Commands</i>	-	-	-	-
PC4. Create and execute basic scripts in Linux/Unix for file management and automation tasks.	-	-	-	-
PC5. Apply Linux commands effectively for navigating the file system, managing processes, and configuring system settings.	-	-	-	-
PC6. Troubleshoot scripting errors and utilize command-line utilities for system administration tasks.	-	-	-	-
<i>Python Programming</i>	-	-	-	-
PC7. Develop Python programs that handle input/output operations, including reading from and writing to files.	-	-	-	-
PC8. Implement control structures like loops, conditions, and functions to solve programming challenges.	-	-	-	-
PC9. Design object-oriented Python programs using classes, inheritance, and encapsulation principles for code organization and reusability	-	-	-	-
NOS Total	50	22	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	NIE/SSC/N1802
NOS Name	Solving biological problems using Python programming
Sector	IT-ITeS
Sub-Sector	
Occupation	Bioinformatics Data Analytics
NSQF Level	4
Credits	3
Version	1.0
Last Reviewed Date	15/03/2024
Next Review Date	15/03/2027
NSQC Clearance Date	15/03/2024

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NIE/SSC/N1803: Analysis and understanding of biomolecular structures using Bioinformatics tools

Description

This NOS delves into the intricacies of biomolecular structures through Bioinformatics tools, covering fundamental concepts such as the physical and chemical properties of amino acids and polypeptides. Students will explore the complexities of protein structures, including non-covalent and covalent interactions, primary and secondary structures, and various structural motifs. The curriculum includes in-depth discussions on protein folding, post-translation modifications, and the relationship between protein sequence, structure, and function.

Scope

The scope covers the following :

- Fundamentals of Biomolecular Structures: Covers amino acid properties, structural motifs, and protein folding principles.
- Protein Structure Determination Techniques: Includes X-ray diffraction, NMR, and electron microscopy methods.
- Exploration of Structural Databases: Introduces databases like PDB, NDB, and tools for structural analysis.
- Prediction, Modeling, and Simulation: Focuses on predicting protein structures, molecular dynamics simulations, and practical applications.

Elements and Performance Criteria

Theoretical Foundations

To be competent, the user/individual on the job must be able to:

PC1. Demonstrate knowledge of amino acid properties and their role in protein structure.

PC2. Explain the principles of secondary structures like alpha helices and beta sheets

PC3. Analyze the factors influencing protein folding and the implications of the Levinthal Paradox

Experimental Techniques

To be competent, the user/individual on the job must be able to:

PC4. Describe the steps involved in protein isolation and purification techniques

PC5. Evaluate different methods of protein structure determination such as X-ray diffraction and NMR spectroscopy.

PC6. Interpret electron density maps and understand the process of structure refinement.

Computational Tools and Databases

To be competent, the user/individual on the job must be able to:

PC7. Use software tools to predict protein secondary structures and analyze structural motifs.

PC8. Demonstrate proficiency in accessing and querying protein databases like PDB and SCOP.

PC9. Apply molecular modeling techniques to simulate protein interactions and dynamics.

Knowledge and Understanding (KU)

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The individual on the job needs to know and understand:

- KU1.** • Protein Structure Fundamentals:
- Understanding the physical and chemical properties of amino acids and their role in forming protein structures.
 - Knowledge of non-covalent and covalent interactions that stabilize protein conformations.
 - Grasping the principles behind primary, secondary, tertiary, and quaternary protein structures.
- KU2.** • Experimental Techniques and Tools:
- Knowledge of experimental techniques like X-ray diffraction and electron microscopy for protein structure determination.
 - Understanding the interpretation of electron density maps and refinement processes for protein structures.
 - Familiarity with structural databases such as PDB, NDB, and tools like DSSP/STRIDE for protein analysis.
- KU3.** • Computational Modeling and Analysis:
- Understanding computational methods for predicting secondary protein structures and identifying structural motifs.
 - Knowledge of molecular modeling techniques, energy minimization algorithms, and simulation methods.
 - Ability to interpret simulation results, analyze protein-protein interactions, and evaluate binding sites.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** • Analytical Skills:
- Ability to analyze complex biomolecular structures using bioinformatics tools and techniques.
 - Capacity to interpret experimental data from X-ray diffraction and electron microscopy for protein structure determination.
 - Capability to critically evaluate protein structures, identify patterns, and analyze structural relationships.
- GS2.** • Problem-Solving Skills:
- Proficiency in applying computational methods for predicting protein structures and performing structural alignments.
 - Skill in troubleshooting and optimizing computational models and simulations for accurate results.
 - Ability to solve complex biological problems related to protein structure, dynamics, and interactions.
- GS3.** • Technical Skills:
- Competence in using bioinformatics software and databases such as PDB, DSSP/STRIDE, and molecular modeling tools.
 - Proficiency in programming languages and scripting for data analysis and manipulation in bioinformatics.
 - Familiarity with scientific writing and communication skills to report and present findings effectively.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Theoretical Foundations</i>	-	-	-	-
PC1. Demonstrate knowledge of amino acid properties and their role in protein structure.	-	-	-	-
PC2. Explain the principles of secondary structures like alpha helices and beta sheets	-	-	-	-
PC3. Analyze the factors influencing protein folding and the implications of the Levinthal Paradox	-	-	-	-
<i>Experimental Techniques</i>	-	-	-	-
PC4. Describe the steps involved in protein isolation and purification techniques	-	-	-	-
PC5. Evaluate different methods of protein structure determination such as X-ray diffraction and NMR spectroscopy.	-	-	-	-
PC6. Interpret electron density maps and understand the process of structure refinement.	-	-	-	-
<i>Computational Tools and Databases</i>	-	-	-	-
PC7. Use software tools to predict protein secondary structures and analyze structural motifs.	-	-	-	-
PC8. Demonstrate proficiency in accessing and querying protein databases like PDB and SCOP.	-	-	-	-
PC9. Apply molecular modeling techniques to simulate protein interactions and dynamics.	-	-	-	-
NOS Total	67	30	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	NIE/SSC/N1803
NOS Name	Analysis and understanding of biomolecular structures using Bioinformatics tools
Sector	IT-ITeS
Sub-Sector	
Occupation	Bioinformatics Data Analytics
NSQF Level	4
Credits	4
Version	1.0
Last Reviewed Date	15/03/2024
Next Review Date	15/03/2027
NSQC Clearance Date	15/03/2024

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NIE/SSC/N1804: Data security law

Description

The course on Data Security Law covers a comprehensive range of topics essential for understanding and implementing effective data protection measures. Students will gain insights into the evolution of data security laws, including key concepts and terminology.

Scope

The scope covers the following :

- **Foundations of Data Security Law:** Covers fundamental concepts, historical evolution, and the importance of data protection.
- **Legal Frameworks and Compliance:** Focuses on major data protection laws, comparative analysis, and compliance challenges.
- **Managing Data Breaches:** Addresses understanding breaches, legal responses, and developing incident plans.
- **Risk Management and Emerging Trends:** Includes risk assessment, compliance strategies, and future developments in data security laws.

Elements and Performance Criteria

Conceptual Understanding

To be competent, the user/individual on the job must be able to:

- PC1.** Demonstrate comprehension of key data security concepts and terminology by accurately defining and explaining terms like encryption, authentication, and data breach.
- PC2.** Identify and describe the historical evolution of data security laws, including landmark legislation and significant events that shaped data protection practices.
- PC3.** Evaluate the importance of data protection in modern society by analyzing case studies, industry reports, and real-world examples to understand the impact of data breaches on businesses and individuals.

Legal Framework Exploration

To be competent, the user/individual on the job must be able to:

- PC4.** Compare and contrast major data protection laws such as GDPR, CCPA, HIPAA, and analyze their scope, requirements, and implications for businesses and consumers.
- PC5.** Critically examine international data security regulations to understand the global landscape of data protection and privacy laws, including differences in approaches and enforcement mechanisms.
- PC6.** Identify and explain jurisdictional differences and compliance challenges related to data security laws, considering factors like cross-border data transfers, data residency requirements, and regulatory oversight.

Practical Application and Response

To be competent, the user/individual on the job must be able to:

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- PC7.** Develop and propose effective strategies for data controllers and processors to comply with data security laws, including data subject rights management, consent mechanisms, and data protection impact assessments.
- PC8.** Create incident response plans and protocols for managing data breaches and security incidents, including steps for detection, containment, mitigation, notification, and recovery.
- PC9.** Evaluate and recommend risk management and compliance strategies, including risk assessment methodologies, technical and organizational measures, and best practices for ensuring data security and regulatory compliance.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.**
- Comprehensive Understanding of Data Security Concepts:
 - Knowledge of fundamental data security concepts and terminology, including encryption methods, authentication mechanisms, data breach prevention strategies, and cybersecurity best practices.
 - Understanding the historical evolution of data security laws, including the development of key legislation such as GDPR, HIPAA, and CCPA, and their impact on data protection practices globally.
 - Awareness of the importance of data protection in modern soc
- KU2.**
- Legal Framework and Compliance Knowledge:
 - Familiarity with major data protection laws and regulations, such as GDPR, CCPA, HIPAA, and their respective legal frameworks, requirements, and enforcement mechanisms.
 - Understanding of international data security regulations and jurisdictional differences, including comparative analysis of data protection laws across different regions and countries.
 - Knowledge of compliance requirements and legal obligations for data controllers and processors, including
- KU3.**
- Practical Application and Response Strategies:
 - Application of legal principles and requirements to real-world scenarios, such as developing compliance strategies, drafting privacy policies, and implementing data protection measures.
 - Understanding of incident response management, including identifying data breaches, assessing the impact, notifying affected parties, and coordinating with regulatory authorities.
 - Knowledge of risk management strategies, including risk assessment methodologies, data

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.**
- Legal Research and Analysis:
 - Ability to conduct in-depth legal research on data security laws, regulations, and compliance requirements.
 - Analytical skills to interpret and analyze complex legal frameworks, including understanding jurisdictional differences and compliance challenges.
 - Capability to synthesize legal information and apply it to practical scenarios, such as assessing the legal implications of data breaches and developing compliance strategies.

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- GS2.** • Communication and Documentation:
- Effective communication skills to convey complex legal concepts, compliance requirements, and risk assessments to various stakeholders within an organization.
 - Proficiency in drafting clear and concise legal documents, including privacy policies, data protection agreements, and incident response plans.
 - Capability to communicate with regulatory authorities, legal teams, and data protection officers regarding compliance issues, data breach notifications, and regulat
- GS3.** • Problem-Solving and Decision-Making:
- Critical thinking skills to analyze data security challenges, identify potential risks and vulnerabilities, and develop proactive risk management strategies.
 - Ability to make informed decisions regarding data protection measures, incident response protocols, and compliance initiatives based on legal requirements and risk assessments.
 - Problem-solving skills to address data breaches, security incidents, and regulatory compliance issues effectively, including dev

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Conceptual Understanding</i>	-	-	-	-
PC1. Demonstrate comprehension of key data security concepts and terminology by accurately defining and explaining terms like encryption, authentication, and data breach.	-	-	-	-
PC2. Identify and describe the historical evolution of data security laws, including landmark legislation and significant events that shaped data protection practices.	-	-	-	-
PC3. Evaluate the importance of data protection in modern society by analyzing case studies, industry reports, and real-world examples to understand the impact of data breaches on businesses and individuals.	-	-	-	-
<i>Legal Framework Exploration</i>	-	-	-	-
PC4. Compare and contrast major data protection laws such as GDPR, CCPA, HIPAA, and analyze their scope, requirements, and implications for businesses and consumers.	-	-	-	-
PC5. Critically examine international data security regulations to understand the global landscape of data protection and privacy laws, including differences in approaches and enforcement mechanisms.	-	-	-	-
PC6. Identify and explain jurisdictional differences and compliance challenges related to data security laws, considering factors like cross-border data transfers, data residency requirements, and regulatory oversight.	-	-	-	-
<i>Practical Application and Response</i>	-	-	-	-
PC7. Develop and propose effective strategies for data controllers and processors to comply with data security laws, including data subject rights management, consent mechanisms, and data protection impact assessments.	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC8. Create incident response plans and protocols for managing data breaches and security incidents, including steps for detection, containment, mitigation, notification, and recovery.	-	-	-	-
PC9. Evaluate and recommend risk management and compliance strategies, including risk assessment methodologies, technical and organizational measures, and best practices for ensuring data security and regulatory compliance.	-	-	-	-
NOS Total	25	11	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	NIE/SSC/N1804
NOS Name	Data security law
Sector	IT-ITeS
Sub-Sector	
Occupation	Bioinformatics Data Analytics
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	15/03/2024
Next Review Date	15/03/2027
NSQC Clearance Date	15/03/2024

Qualification Pack

NIE/SSC/N1810: Implementation of project on bioinformatics

Description

This NOS defines the skills required for implementing a bioinformatics project, focusing on computational tools, programming techniques, and biological data analysis. It covers key areas such as sequence analysis, structural modeling, and data visualization while ensuring accuracy in biological interpretations. Effective collaboration, problem-solving, and staying updated with emerging technologies and best practices are essential aspects of this standard.

Scope

The scope covers the following :

- The scope of this NOS includes the implementation of bioinformatics projects using computational tools, algorithms, and programming techniques. It covers sequence analysis, structural modeling, biological data visualization, and statistical analysis to derive meaningful insights. The NOS also emphasizes the integration of databases, machine learning applications, and workflow automation for efficient data processing. Additionally, it highlights the importance of accuracy, collaboration, and adherence to best practices in bioinformatics research and development.

Elements and Performance Criteria

Execution of Bioinformatics Project

To be competent, the user/individual on the job must be able to:

- PC1.** Understand fundamental bioinformatics concepts, including computational tools, biological databases, and data analysis techniques.
- PC2.** Develop and implement bioinformatics algorithms for sequence analysis, structural modeling, and biological data interpretation.
- PC3.** Utilize data visualization techniques to analyze and present biological findings effectively.
- PC4.** Collaborate with interdisciplinary teams and stay updated with advancements in bioinformatics tools, programming languages, and best practices.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Fundamental concepts of bioinformatics, including biological databases, sequence alignment, and molecular modeling.
- KU2.** Computational tools and algorithms used for genomic data analysis, protein structure prediction, and functional annotation.
- KU3.** Data visualization techniques for interpreting and presenting bioinformatics results effectively.

Generic Skills (GS)

User/individual on the job needs to know how to:

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- GS1.** Develop expertise in using bioinformatics tools and software for sequence analysis, molecular modeling, and genome annotation. Acquire the ability to handle large biological datasets and perform computational simulations effectively.
- GS2.** Cultivate strong analytical skills to interpret genomic and proteomic data, apply statistical methods for biological research, and troubleshoot computational challenges in bioinformatics projects.
- GS3.** Enhance the ability to effectively communicate bioinformatics concepts and research findings. Prepare clear and detailed documentation, including data reports, algorithm descriptions, and result interpretations, to facilitate collaboration with researchers and stakeholders.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Execution of Bioinformatics Project</i>	-	-	-	-
PC1. Understand fundamental bioinformatics concepts, including computational tools, biological databases, and data analysis techniques.	-	-	-	-
PC2. Develop and implement bioinformatics algorithms for sequence analysis, structural modeling, and biological data interpretation.	-	-	-	-
PC3. Utilize data visualization techniques to analyze and present biological findings effectively.	-	-	-	-
PC4. Collaborate with interdisciplinary teams and stay updated with advancements in bioinformatics tools, programming languages, and best practices.	-	-	-	-
NOS Total	-	-	160	-

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National Occupational Standards (NOS) Parameters

NOS Code	NIE/SSC/N1810
NOS Name	Implementation of project on bioinformatics
Sector	IT-ITeS
Sub-Sector	
Occupation	Bioinformatics Data Analytics
NSQF Level	4
Credits	1
Version	1.0
Last Reviewed Date	15/03/2024
Next Review Date	15/03/2027
NSQC Clearance Date	15/03/2024

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DGT/VSQ/N0102: Employability Skills (60 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Career Development & Goal Setting
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

- PC1.** identify employability skills required for jobs in various industries
- PC2.** identify and explore learning and employability portals

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

- PC3.** recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.
- PC4.** follow environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

- PC5.** recognize the significance of 21st Century Skills for employment
- PC6.** practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life

Basic English Skills

To be competent, the user/individual on the job must be able to:

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- PC7.** use basic English for everyday conversation in different contexts, in person and over the telephone
- PC8.** read and understand routine information, notes, instructions, mails, letters etc. written in English
- PC9.** write short messages, notes, letters, e-mails etc. in English

Career Development & Goal Setting

To be competent, the user/individual on the job must be able to:

- PC10.** understand the difference between job and career
- PC11.** prepare a career development plan with short- and long-term goals, based on aptitude

Communication Skills

To be competent, the user/individual on the job must be able to:

- PC12.** follow verbal and non-verbal communication etiquette and active listening techniques in various settings
- PC13.** work collaboratively with others in a team

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

- PC14.** communicate and behave appropriately with all genders and PwD
- PC15.** escalate any issues related to sexual harassment at workplace according to POSH Act

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

- PC16.** select financial institutions, products and services as per requirement
- PC17.** carry out offline and online financial transactions, safely and securely
- PC18.** identify common components of salary and compute income, expenses, taxes, investments etc
- PC19.** identify relevant rights and laws and use legal aids to fight against legal exploitation

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

- PC20.** operate digital devices and carry out basic internet operations securely and safely
- PC21.** use e-mail and social media platforms and virtual collaboration tools to work effectively
- PC22.** use basic features of word processor, spreadsheets, and presentations

Entrepreneurship

To be competent, the user/individual on the job must be able to:

- PC23.** identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research
- PC24.** develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion
- PC25.** identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity

Customer Service

To be competent, the user/individual on the job must be able to:

- PC26.** identify different types of customers
- PC27.** identify and respond to customer requests and needs in a professional manner.

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PC28. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC29. create a professional Curriculum vitae (Résumé)

PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively

PC31. apply to identified job openings using offline /online methods as per requirement

PC32. answer questions politely, with clarity and confidence, during recruitment and selection

PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills and different learning and employability related portals

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use English language for effective verbal (face to face and telephonic) and written communication in formal and informal set up

KU6. importance of career development and setting long- and short-term goals

KU7. about effective communication

KU8. POSH Act

KU9. Gender sensitivity and inclusivity

KU10. different types of financial institutes, products, and services

KU11. how to compute income and expenditure

KU12. importance of maintaining safety and security in offline and online financial transactions

KU13. different legal rights and laws

KU14. different types of digital devices and the procedure to operate them safely and securely

KU15. how to create and operate an e- mail account and use applications such as word processors, spreadsheets etc.

KU16. how to identify business opportunities

KU17. types and needs of customers

KU18. how to apply for a job and prepare for an interview

KU19. apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read and write different types of documents/instructions/correspondence

GS2. communicate effectively using appropriate language in formal and informal settings

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- GS3.** behave politely and appropriately with all
- GS4.** how to work in a virtual mode
- GS5.** perform calculations efficiently
- GS6.** solve problems effectively
- GS7.** pay attention to details
- GS8.** manage time efficiently
- GS9.** maintain hygiene and sanitization to avoid infection

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. identify employability skills required for jobs in various industries	-	-	-	-
PC2. identify and explore learning and employability portals	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC3. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.	-	-	-	-
PC4. follow environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	2	4	-	-
PC5. recognize the significance of 21st Century Skills for employment	-	-	-	-
PC6. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC7. use basic English for everyday conversation in different contexts, in person and over the telephone	-	-	-	-
PC8. read and understand routine information, notes, instructions, mails, letters etc. written in English	-	-	-	-
PC9. write short messages, notes, letters, e-mails etc. in English	-	-	-	-
<i>Career Development & Goal Setting</i>	1	2	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. understand the difference between job and career	-	-	-	-
PC11. prepare a career development plan with short- and long-term goals, based on aptitude	-	-	-	-
<i>Communication Skills</i>	2	2	-	-
PC12. follow verbal and non-verbal communication etiquette and active listening techniques in various settings	-	-	-	-
PC13. work collaboratively with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	2	-	-
PC14. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC15. escalate any issues related to sexual harassment at workplace according to POSH Act	-	-	-	-
<i>Financial and Legal Literacy</i>	2	3	-	-
PC16. select financial institutions, products and services as per requirement	-	-	-	-
PC17. carry out offline and online financial transactions, safely and securely	-	-	-	-
PC18. identify common components of salary and compute income, expenses, taxes, investments etc	-	-	-	-
PC19. identify relevant rights and laws and use legal aids to fight against legal exploitation	-	-	-	-
<i>Essential Digital Skills</i>	3	4	-	-
PC20. operate digital devices and carry out basic internet operations securely and safely	-	-	-	-
PC21. use e- mail and social media platforms and virtual collaboration tools to work effectively	-	-	-	-
PC22. use basic features of word processor, spreadsheets, and presentations	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Entrepreneurship</i>	2	3	-	-
PC23. identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research	-	-	-	-
PC24. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion	-	-	-	-
PC25. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity	-	-	-	-
<i>Customer Service</i>	1	2	-	-
PC26. identify different types of customers	-	-	-	-
PC27. identify and respond to customer requests and needs in a professional manner.	-	-	-	-
PC28. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	2	3	-	-
PC29. create a professional Curriculum vitae (Résumé)	-	-	-	-
PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively	-	-	-	-
PC31. apply to identified job openings using offline /online methods as per requirement	-	-	-	-
PC32. answer questions politely, with clarity and confidence, during recruitment and selection	-	-	-	-
PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements	-	-	-	-
NOS Total	20	30	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0102
NOS Name	Employability Skills (60 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	12/03/2026
Next Review Date	12/03/2029
NSQC Clearance Date	12/03/2026

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

- Assessment of the qualification evaluates candidates to ascertain that they can integrate knowledge, skills and values for carrying out relevant tasks as per the
- defined learning outcomes and assessment criteria.
- The underlying principle of assessment is fairness and transparency. The evidence of the outcomes and assessment criteria. competence acquired by the candidate • can be obtained by conducting Theory (Online), Practical assessment, Internal assessment, Project/Presentation/ Assignment, Major Project. The emphasis is on the
- practical demonstration of skills & knowledge gained by the candidate through the training. Each OUTCOME is assessed & marked separately. A candidate is
- required to pass all OUTCOMES individually based on the passing criteria.
- About Examination Pattern:
 1. The question papers for the theory and practical exams are set by the Examination wing (assessor) of NIELIT HQS.

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- 2. The assessor assigns roll number
- 3. The assessor carries out theory online assessments through remote proctoring methodology. Theory examination would be conducted online and the paper comprise of MCQ. Conduct of assessment are through trained proctors. Once the test begins, remote proctors have full access to candidate's video feeds and computer screens. Proctors authenticate the candidate based on registration details, pre-test image captured and I- card in possession of the candidate. Proctors can chat with candidates or give warnings to candidates. Proctors can also take screenshots, terminate a specific user's test session, or re-authenticate candidates based on video feeds.
- 4. An External Examiner/ Observer may be deployed including NIELIT officials for evaluation of Practical examination/ internal assessment / Project/ Presentation/. Major Project (if applicable) would be evaluated preferably by external/ subject expert including NIELIT officials.
- 5. Pass percentage would be 50% marks in each component.
- 6. Candidates may apply for re-examination within the validity of registration (only in the assessment component in which the candidate failed).
- 7. For re-examination prescribed examination fee is required to be paid by the candidate only for the assessment component in which the candidate wants to reappear.
- 8. There would be no exemption for any paper/module for candidates having similar qualifications or skills.
- 9. The examination will be conducted in English language only.
- Quality assurance activities: A pool of questions is created by a subject matter expert and moderated by other SME. Test rules are set beforehand. Random set of questions which are according to syllabus appears which may differ from candidate to candidate. Confidentiality and impartiality are maintained during all the examination and evaluation processes.

Minimum Aggregate Passing % at QP Level : 50

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

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National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
NIE/SSC/N1801.Algorithmic perspective of Bioinformatics and introduction to Bioinformatics tools.	50	23	-	-	73	15
NIE/SSC/N1802.Solving biological problems using Python programming	50	22	-	-	72	14
NIE/SSC/N1803.Analysis and understanding of biomolecular structures using Bioinformatics tools	67	30	-	-	97	20
NIE/SSC/N1804.Data security law	25	11	-	-	36	9
NIE/SSC/N1810.Implementation of project on bioinformatics	-	-	160	-	160	32
DGT/VSQ/N0102.Employability Skills (60 Hours)	20	30	-	-	50	10
Total	212	116	160	-	488	100

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Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training
AA	Assessment Agency
AB	Awarding Body
ISCO	International Standard Classification of Occupations
NCO	National Classification of Occupations
NCrF	National Credit Framework
NOS	National Occupational Standard(s)
NQR	National Qualification Register
NSQF	National Skills Qualifications Framework
OJT	On the Job Training
AA	Assessment Agency
AB	Awarding Body
NCO	National Classification of Occupations
NCrF	National Credit Framework
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NCrF	National Credit Framework
NOS	National Occupational Standard(s)
NQR	National Qualification Register
NSQF	National Skills Qualifications Framework
OJT	On the Job Training

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Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.
Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.

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Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.
National Occupational Standar	NOS defines the measurable performance outcomes required from an individual engaged in a particular task. They list down what an individual performing that task should know and also do.
Qualification	A formal outcome of an assessment and validation process which is obtained when a competent body determines that an individual has achieved learning outcomes to given standards
Qualification File	A Qualification File is a template designed to capture necessary information of a Qualification from the perspective of NSQF compliance. The Qualification File will be normally submitted by the awarding body for the qualification.
Sector	A grouping of professional activities on the basis of their main economic function, product, service or technology.
Long Term Training	Long-term skilling means any vocational training program undertaken for a year and above. https://ncvet.gov.in/sites/default/files/NCVET.pdf
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